Gitlab hosted link: https://adrianhamel.github.io/FinalProject-380/public/

For fun performance checking, take a look at: https://adrianhamel.github.io/FinalProject-380/lighthouse-results/lighthouse-results-13.html

Paper below in text form

Impact of Blizzard Battle.Net on the 1996s PC gaming industry.

I. Introduction

In the early 1990s, online gaming was fragmented, inconsistent, and extremely difficult for the average player to access. Connecting to another person required technical knowledge, modem-to-modem dialing, and manually entering IP addresses, or relying on IPX tunneling programs. Even when the connection succeeded, the experience varied widely, as each game used its own peer to peer connections. In the early 90s, gaming related social interaction was minimal and often limited to software like the internet relay chat. Paid services attempted to fix this but ended up adding financial barriers, separating gaming communities. Battle.net transformed the online gaming space by removing these barriers such as cost, technical barriers in difficulty connecting, and centralizing communication between the larger player base, revolutionizing the industry today.

Battle.net's contribution to online gaming went far beyond simplifying and fixing these difficulties. With multiplayer connections, it reshaped the way players interacted, communicated, and played. By fixing the connection challenges, removing the subscription fees and paid services, and supporting an online community, Blizzard created a new community where everyone could play. Blizzard entertainment with Battle.net revolutionized gaming with its technological innovations through creating online social worlds, the rise of clans and communities, and the creation of new peer to peer connections that had a lasting impact on the industry as a whole.

II. Online Gaming in the 1990s

Before the Battle.net platform was created, online multiplayer gaming depended heavily on direct peer to peer connections. Gamers used dial up modems that were slow, prone to failure, and were often disrupted by another person wanting to use the phone line. Something simple like a family member picking up the phone would disconnect a player from a match. To start initiating a connection, gamers had to exchange IP addresses and phone numbers to start. To do this, they would use communication channels like IRC, gaming forums or magazines to find other groups hosting games.

IPX tunneling programs such as Kali or iDOOM offered temporary solutions to the connection problem by emulating a LAN connection over the internet. The IPX standard to this day has been retired and games that use it can no longer function without workarounds. While these programs solved some issues, they required users to configure ports, rewrite network drivers, adjust packet and other network settings, and troubleshoot issues without on their own without documentation. On top of the financial costs, the technicality of using them still kept the online playerbase limited. According to sources like reddit, games like starcraft used Kali but this brought a lot of issues with lag. The game had to use a special patch to make the lag playable.

Another financial gap came from TEN and Heat.net. These software made gaming more accessible with marketing for matchmaking rooms and social channels, but it was locked behind a paywall. Since gaming was not as mainstream as it is today, some people were discouraged from paying the cost as it did not prove itself yet. Online multiplayer across different game titles also lacked consistency. One game might use modem dialing while another might require LAN mode and others used proprietary online connection systems that did not guarantee to function. Since there were no unified chatrooms or built in community forums, players were left figuring out the different solutions on their own. This created the fragmented landscape of online gaming that felt experimental, unstable, and heavily dependent on each person figuring out how to setup each game.

III. Battle.net Launches

Blizzard designed Battle.net as a free, simple, community-oriented and driven platform. On November 30, 1996, Battle.net launched with the Diablo, giving players an option to log into an online ecosystem with one click. This removed the need for figuring out IPs and modem numbers, configuring ports, network driver software, or running additional software, by placing all multiplayer features inside the game itself.

Battle.net introduced chat lobbies, game specific rooms, clan spaces, and a match finder for active games. This unification of matchmaking and communication in one place made gaming easier and accessible. The system used a centralized login servers system, item and character verification for Diablo to reduce cheating and counter hacks and dupes, and a lobby to game session feature that gave players a consistent experience in matchmaking. The platform grew quickly with more than one million users had logged into Battle.net by 1997. The platform expanded with support for StarCraft, Brood War, and Warcraft II Battle.net Edition by 1998. Since StarCraft was massively popular, Battle.net also became more popular since it was competitive and had an explosion of clans that utilized its chat features. Because of this, Battle.net turned into a whole new social ecosystem.

IV. Social Worlds and Community Forums

Battle.net's had public chat rooms that worked like IRC except built in so gamers did not need to run extra software or another IRC chat. These built in chats were social hubs for the individual games. They were public and gamers were able to talk and build online relationships, which is what really pushed this platform forward. These channels often had hundreds of users, creating a sense of population and activity that players did not get to experience with isolated peer to peer connections like IRC offered. These chats were flooded with ASCII art, random spam messages, item trades, clan advertisements, information on hacks and dupes, and casual conversations creating a vibrant culture. Gamers developed their online identities, with unique usernames and clan affiliations. Because of how big the social community was, reaching gloabal scales, some players logged into Battle.net to simply talk for hours, treating the platform as a virtual hangout spot like modern day social media chats.

Clan culture became one of the strongest forces in Battle.net's social environment and the main reason for its popularity. Clans were organized with a dedicated hierarchy, including leaders, officers, trainees, and even practice teams. Many clans required tryouts or skill tests to join, which were once again made simple due to Battle.nets matchmaking multiplayer abilities. Clan wars between groups created competitions and fueled rivalries.

Clan tags became symbols of belonging and a players status, and players took pride in representing their groups across matches and chatrooms. These clan tags were put in their names and were often acronyms of which clan they represent. It made their groups more like a family. Another benefit of the social community created was StarCraft's map editor. This added another layer to the community experience. Players created custom maps then shared them in chatrooms. This was used to create new ways to play a game, creating new games within the game itself. These early custom community maps became games in their own rights in modern games today. This also became the start of new gaming genres like the first Tower Defense maps, RPG styled experiences, and early MOBA ideas came from hobbyist creators sharing custom maps on Battle.net and eventually became big mainstream games. These maps spread quickly due to the social nature of the platform, shaping future game design and influencing major gaming genres in the early 2000s.

Although Battle.net being free and open was a great things, there were some downsides to true online freedom. Spam bots flooded channels, automated scripts were used to overtake conversations, and harassment was common. To fight this, gamers often used unofficial moderation bots to manage channels by creating chat filters, creating a community moderated atmosphere where control shifted to creators, clans, and social groups. This mixture of freedom, creativity, and chaos made Battle.net feel like a digital homes where gamers shaped the environment themselves and made it their own.

V. The Culture of Hacks, Duping, and Exploitation

Hacking and duping became major cultural elements of the early Battle.net. In Diablo, network packet manipulation allowed players to duplicate rare items by tricking the game that an item existed twice. These dupes circulated widely, forming unofficial economies where players traded cloned and the cheats themselves items for reputation, influence and money. Some players built entire identities around collecting and distributing dupes and cheats. Third party editing tools became common, allowing players to alter character stats, generate impossible items, or inject game packets. Some hacks even caused opponents to crash or lose control of their game sessions. Hackers shared their tools openly in public channels, which spread rapidly through the community. This created a culture where hacks were not just technical exploits but almost social status symbols. Players who discovered or distributed hacks gained recognition. Some clans specialized in hacking, while others formed anti hack groups to fight them as the cheats made the games and competitive matches unfair. Battle.net became a battleground between honest players, exploit users, and the developers. Blizzard then introduced server-side verification and later the Warden anti cheat system to fight these cheats, but exploit creators continued to evolve their methods. This battle between Blizzard and hackers became part of Battle.net's identity, shaping how future anti cheat systems would operate across the industry. Other companies today like steam or riot games have their own full system kernel level anti-cheats to detect the third party software's or unsafe software modifications. Games were validated before they ran to remove any modifications.

VI. Economic and Industry Impacts

Battle.net reshaped online gaming economics by making multiplayer free. Paid network services struggled to compete, and subscription based systems declined rapidly, and eventually died out. Battle.net demonstrated that free online multiplayer could attract millions of users and become a massive selling point for games and their foothold in the industry. During the 2000s, major platforms like Xbox Live, Steam, and PlayStation Network built on ideas that Battle.net introduced. Features like friends lists, chatrooms, matchmaking, and centralized accounts were influenced directly by Blizzard's model. These were for the most part even kept free aside from Xbox, keeping the online features behind a paywell that worked because of their exclusivity.

Esports expanded because of Battle.net's impact all around the world. South Korea for example, StarCraft became a competitive sensation where gamers trained on Battle.net ladders, clans provided organized environments and events, and televised matches helped turn gaming into a spectator event. Battle.net supported the early esports culture by providing the matchmaking structure and large player pools.

VII. Legacy and Modernization

By the 2010s, Battle.net evolved into the Blizzard Launcher, adding patch tools, voice chat, cross game social systems, and digital distribution of their titles. Although it became more controlled, the core ideas from 1996 continued to influence design decisions. Modern platforms such as Discord, Steam, Xbox Live, and PSN continue to use systems Battle.net pioneered with streaming services like Twitch or YouTube were built in to provide quick access streaming. The centralized accounts, social environments and groups, and integrated multiplayer features remain a standard in gaming. Battle.net proved that online games were not just about the gameplay but also about belonging to a community.

VIII. Conclusion

The 1990s were an important decade for online gaming, shaped by technical challenges and barriers for online connectivity. Battle.net unified these multiplayer systems, removed financial and technical barriers, and created strong communities. It influenced digital identities, online cultures, the creation of clans, and early hacking communities. Battle.net created the foundation for global, social, creative, and competitive online gaming that shaped modern platforms and continues to influence the modern gaming culture today.

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